

Strength of Materials II



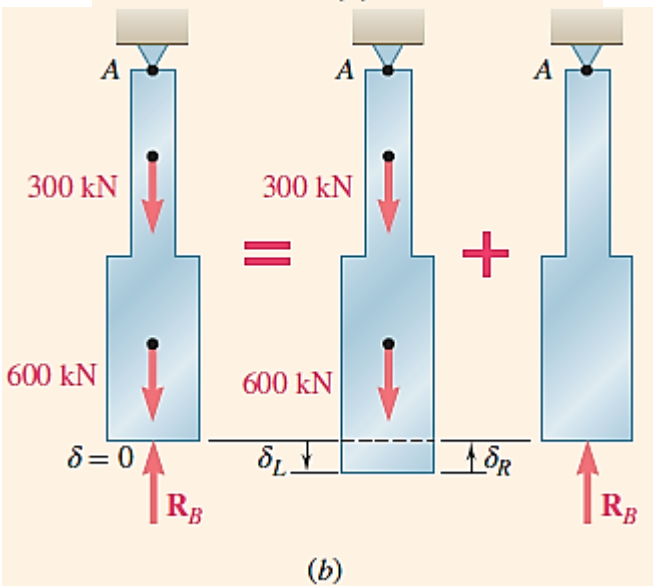
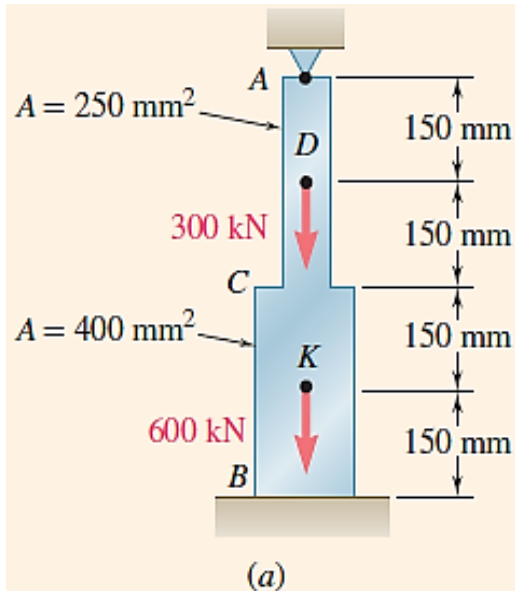
Chapter Five: Static Indeterminate Structures

MEng3111



Inst. : Dawit M.

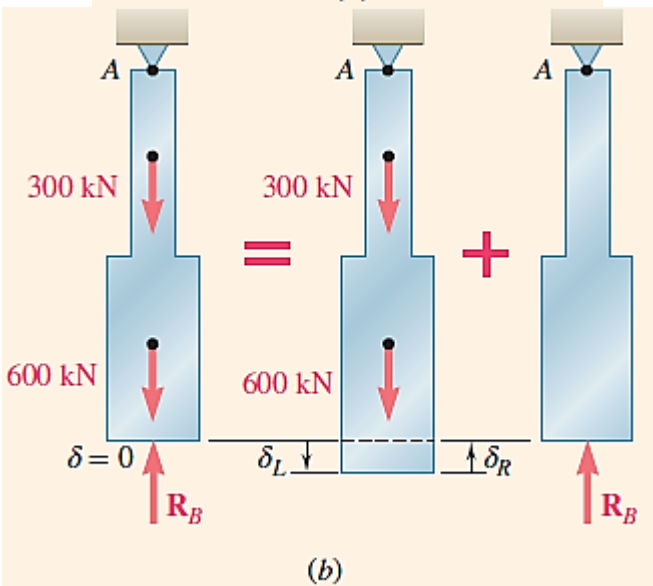
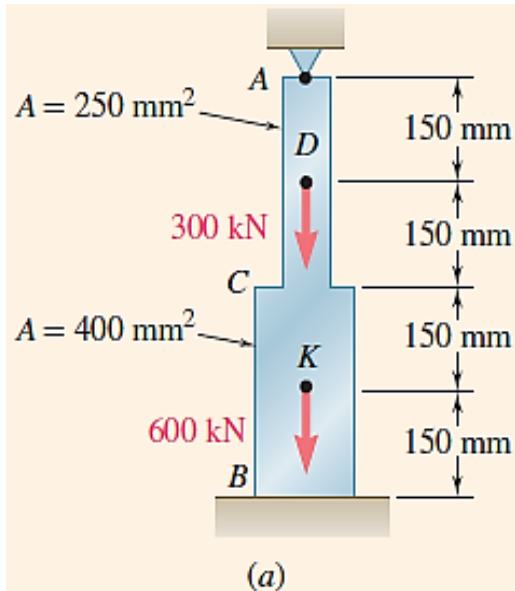
Static Indeterminacy-Axial Loading Case



- Structures for which internal forces and reactions cannot be determined from statics alone are said to be *statically indeterminate*.
- A structure will be statically indeterminate whenever it is held by more supports than are required to maintain its equilibrium.
- Redundant reactions** are replaced with unknown loads which along with the other loads must produce compatible deformations.
- Deformations due to actual loads and redundant reactions are determined separately and then added or *superposed*.

$$\delta = \delta_L + \delta_R = 0$$

Example



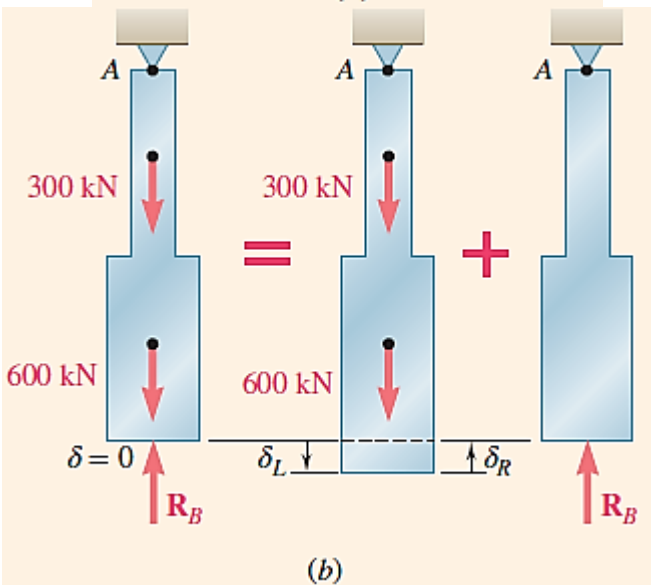
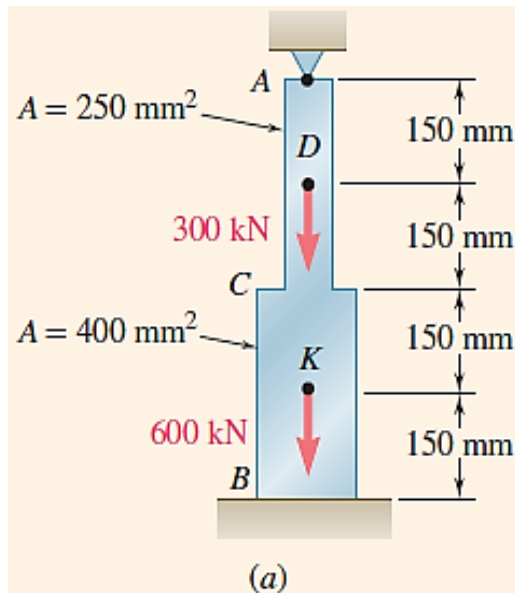
Determine the reactions at A and B for the steel bar and loading shown, assuming a close fit at both supports before the loads are applied.

SOLUTION:

- Consider the reaction at B as redundant, release the bar from that support, and solve for the displacement at B due to the applied loads.
- Solve for the displacement at B due to the redundant reaction at B.
- Require that the displacements due to the loads and due to the redundant reaction be compatible, i.e., require that their sum be zero.
- Solve for the reaction at A due to applied loads and the reaction found at B.

Example

SOLUTION:



- Solve for the displacement at B due to the applied loads with the redundant constraint released,

$$P_1 = 0 \quad P_2 = P_3 = 600 \times 10^3 \text{ N} \quad P_4 = 900 \times 10^3 \text{ N}$$

$$A_1 = A_2 = 400 \times 10^{-6} \text{ m}^2 \quad A_3 = A_4 = 250 \times 10^{-6} \text{ m}^2$$

$$L_1 = L_2 = L_3 = L_4 = 0.150 \text{ m}$$

$$\delta_L = \sum_i \frac{P_i L_i}{A_i E_i} = \frac{1.125 \times 10^9}{E}$$

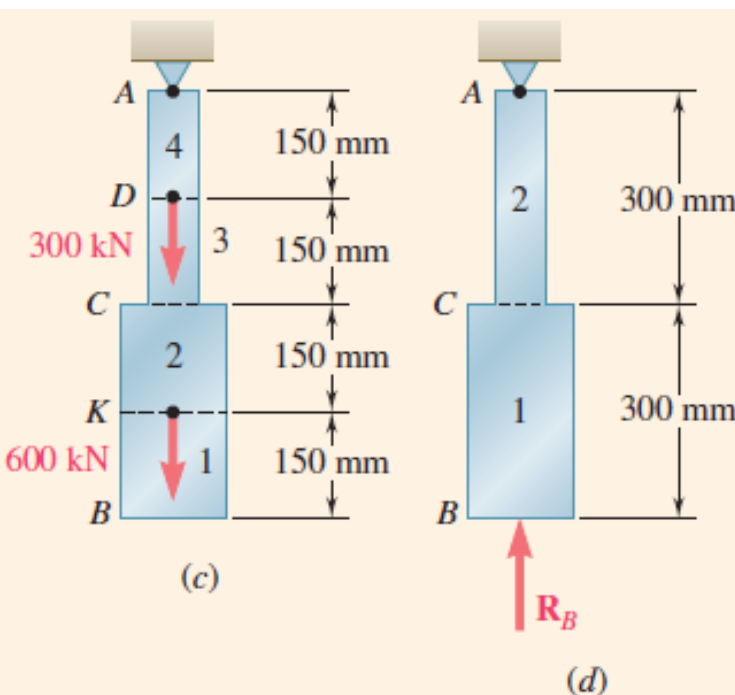
- Solve for the displacement at B due to the redundant constraint,

$$P_1 = P_2 = -R_B$$

$$A_1 = 400 \times 10^{-6} \text{ m}^2 \quad A_2 = 250 \times 10^{-6} \text{ m}^2$$

$$L_1 = L_2 = 0.300 \text{ m}$$

$$\delta_R = \sum_i \frac{P_i L_i}{A_i E_i} = -\frac{(1.95 \times 10^3) R_B}{E}$$



- Require that the displacements due to the loads and due to the redundant reaction be compatible,

$$\delta = \delta_L + \delta_R = 0$$

$$\delta = \frac{1.125 \times 10^9}{E} - \frac{(1.95 \times 10^3) R_B}{E} = 0$$

$$R_B = 577 \times 10^3 \text{ N} = 577 \text{ kN}$$

- Find the reaction at A due to the loads and the reaction at B

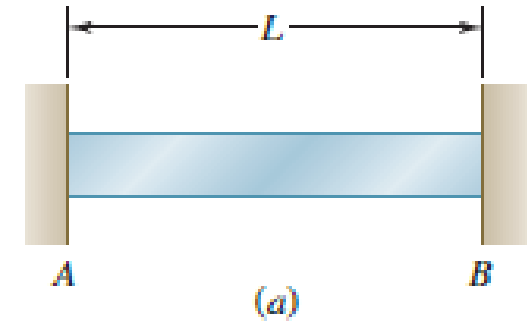
$$\sum F_y = 0 = R_A - 300 \text{ kN} - 600 \text{ kN} + 577 \text{ kN}$$

$$R_A = 323 \text{ kN}$$

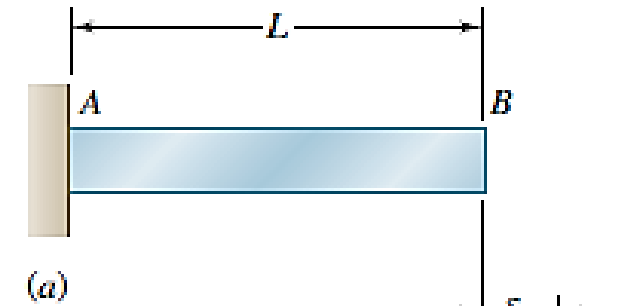
$$R_A = 323 \text{ kN}$$

$$R_B = 577 \text{ kN}$$

Thermal Stresses



- A temperature change results in a change in length or *thermal strain*. There is no stress associated with the thermal strain unless the elongation is restrained by the supports.



- Treat the additional support as redundant and apply the principle of superposition.

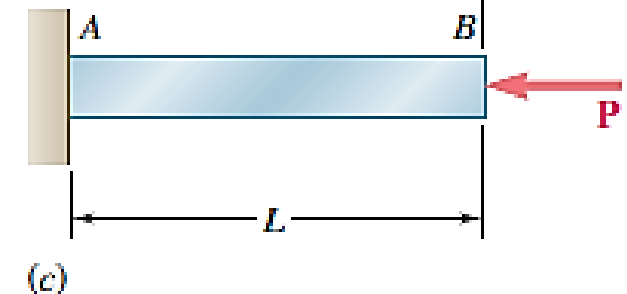
$$\delta_T = \alpha(\Delta T)L$$

$$\delta_P = \frac{PL}{AE}$$

α = thermal expansion coef.



- The thermal deformation and the deformation from the redundant support must be compatible.



$$\delta = \delta_T + \delta_P = 0$$

$$\delta = \delta_T + \delta_P = 0$$

$$\alpha(\Delta T)L + \frac{PL}{AE} = 0$$

$$P = -AE\alpha(\Delta T)$$

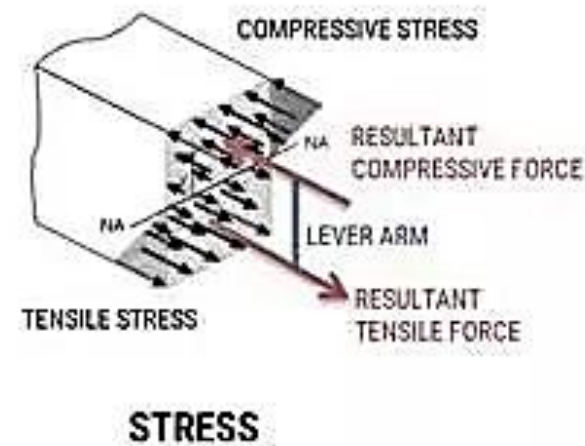
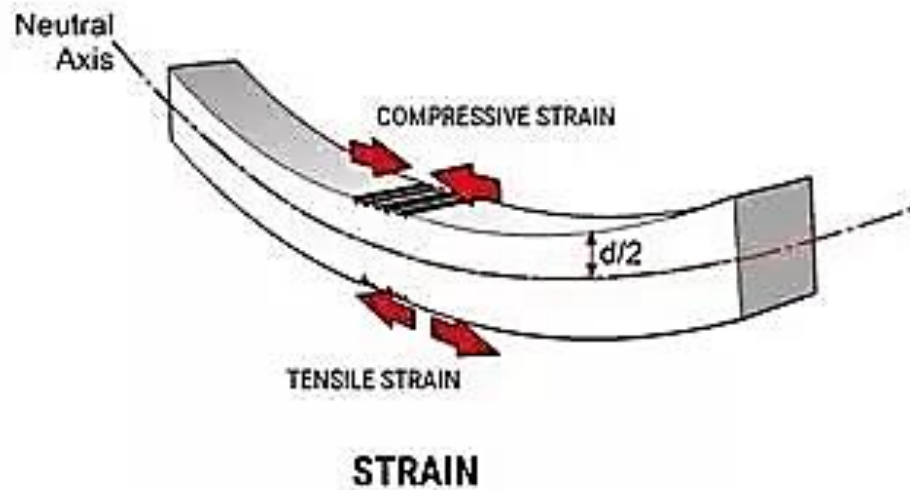
$$\sigma = \frac{P}{A} = -E\alpha(\Delta T)$$

HOMEWORK

- Concept Application 2.2
- Concept Application 2.3
- Concept Application 3.5
- Sample Problem 3.3
- Sample Problem 3.4
- Sample Problem 3.5

Beam-Revision

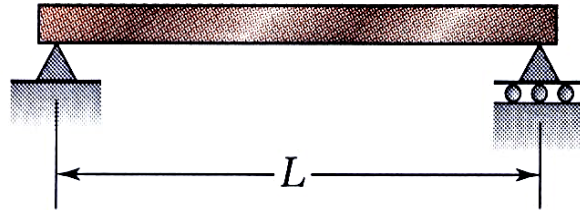
When a beam is loaded by forces or couples, stresses and strains are created throughout the interior of the beam.



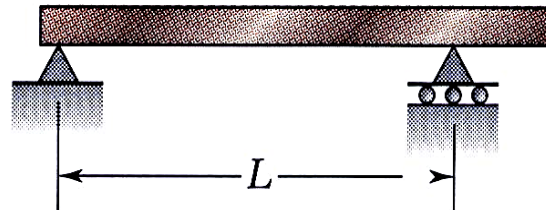
To determine these stresses and strains, the internal forces and internal couples that act on the cross sections of the beam must be found.

Classification of Beam Supports

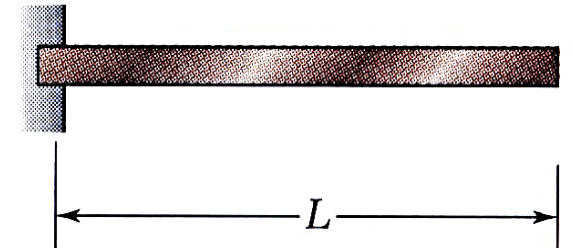
Statically
Determinate
Beams



(a) Simply supported beam



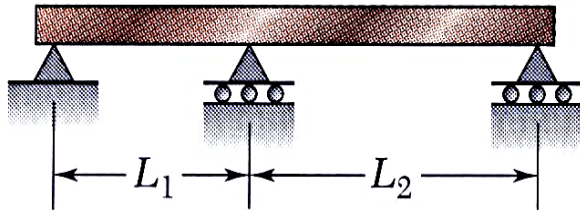
(b) Overhanging beam



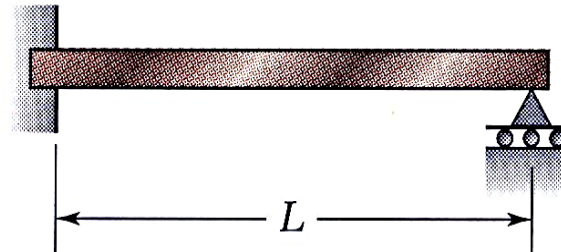
(c) Cantilever beam

L= Span

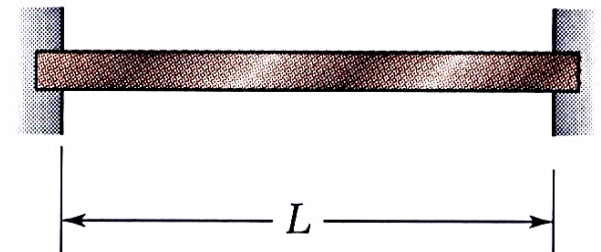
Statically
Indeterminate
Beams



(d) Continuous beam



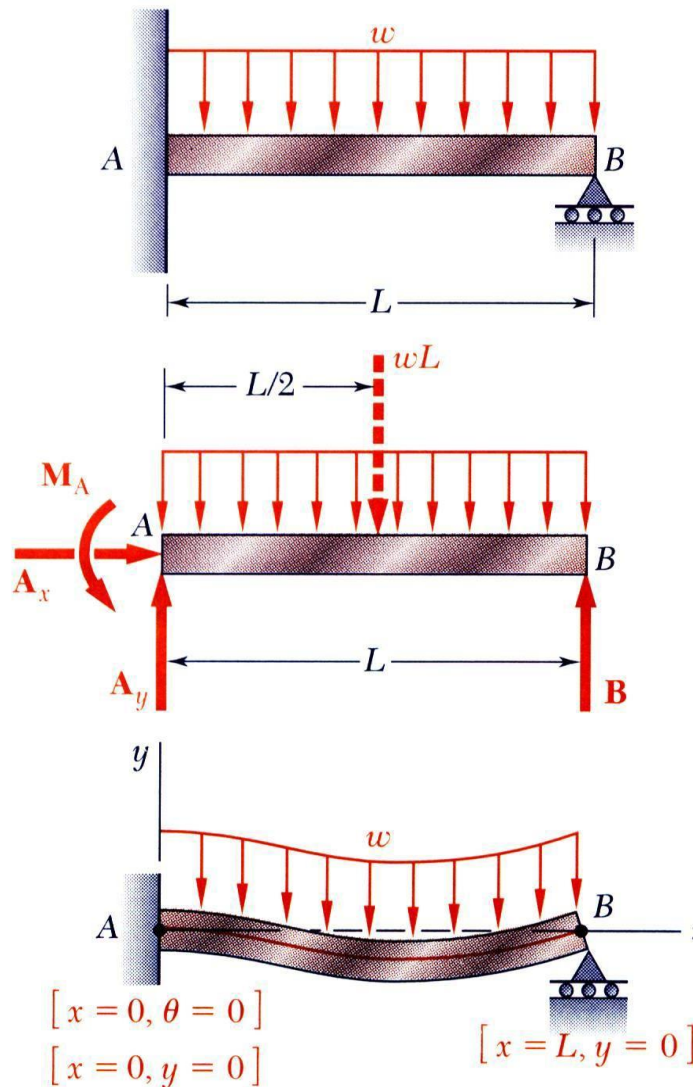
(e) Beam fixed at one end
and simply supported
at the other end



(f) Fixed beam

Total of only three unknowns and can be determined by the methods of statics. Such beams are said to be **statically determinate**.

Statically Indeterminate Beams



- Consider beam with fixed support at A and roller support at B .
- From free-body diagram, note that there are four unknown reaction components.
- Conditions for static equilibrium yield

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_A = 0$$

The beam is statically indeterminate.

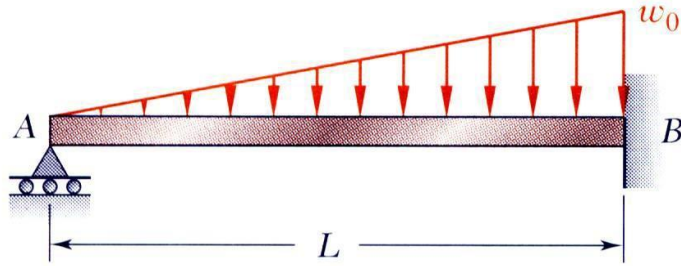
- Also have the beam deflection equation,

$$EI y = \int_0^x dx \int_0^x M(x) dx + C_1 x + C_2$$

which introduces two unknowns but provides three additional equations from the boundary conditions:

$$\text{At } x = 0, \theta = 0 \quad y = 0 \quad \text{At } x = L, y = 0$$

Sample Problem 9.3

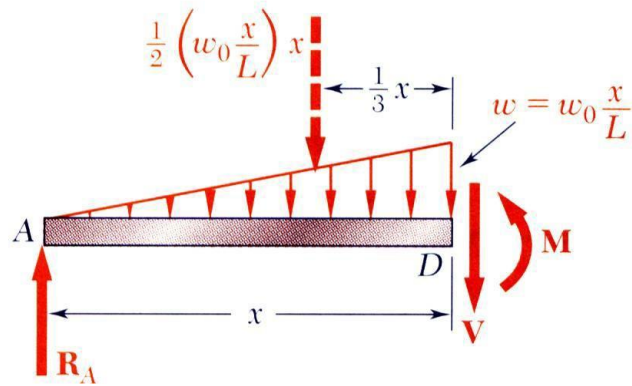


For the uniform beam, determine the reaction at A , derive the equation for the elastic curve, and determine the slope at A . (Note that the beam is statically indeterminate to the first degree)

SOLUTION:

- Develop the differential equation for the elastic curve (will be functionally dependent on the reaction at A).
- Integrate twice and apply boundary conditions to solve for reaction at A and to obtain the elastic curve.
- Evaluate the slope at A .

Sample Problem 9.3



- Consider moment acting at section D ,

$$\sum M_D = 0$$

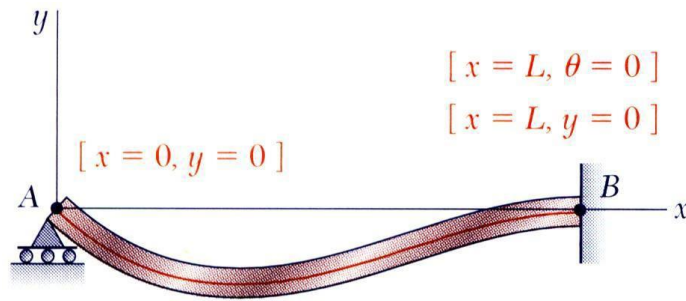
$$R_A x - \frac{1}{2} \left(\frac{w_0 x^2}{L} \right) \frac{x}{3} - M = 0$$

$$M = R_A x - \frac{w_0 x^3}{6L}$$

- The differential equation for the elastic curve,

$$EI \frac{d^2 y}{dx^2} = M = R_A x - \frac{w_0 x^3}{6L}$$

Sample Problem 9.3



$$EI \frac{d^2 y}{dx^2} = M = R_A x - \frac{w_0 x^2}{2}$$

- Integrate twice

$$EI \frac{dy}{dx} = EI \theta = \frac{1}{2} R_A x^2 - \frac{w_0 x^3}{6} + C_1$$

$$EI y = \frac{1}{6} R_A x^3 - \frac{w_0 x^4}{24} + C_1 x + C_2$$

- Apply boundary conditions:

at $x=0, y=0$: $C_2 = 0$

at $x=L, \theta=0$: $\frac{1}{2} R_A L^2 - \frac{w_0 L^3}{6} + C_1 = 0$

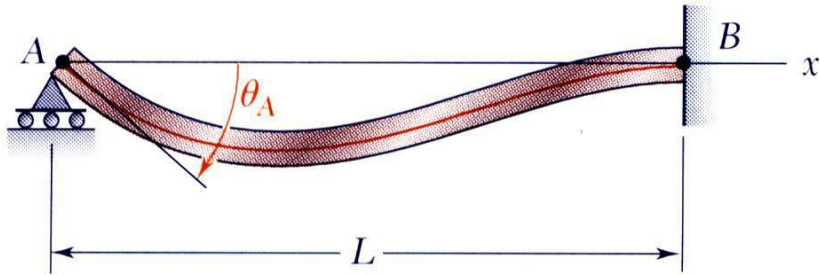
at $x=L, y=0$: $\frac{1}{6} R_A L^3 - \frac{w_0 L^4}{24} + C_1 L + C_2 = 0$

- Solve for reaction at A

$$R_A = \frac{1}{10} w_0 L \uparrow$$

$$C_1 = \frac{1}{120} w_0 L^3$$

Sample Problem 9.3



- Substitute for C_1 , C_2 , and R_A in the elastic curve equation,

$$EI y = \frac{1}{6} \left(\frac{1}{10} w_0 L \right) x^3 - \frac{w_0 x^5}{120 L} - \left(\frac{1}{120} w_0 L^3 \right) x$$

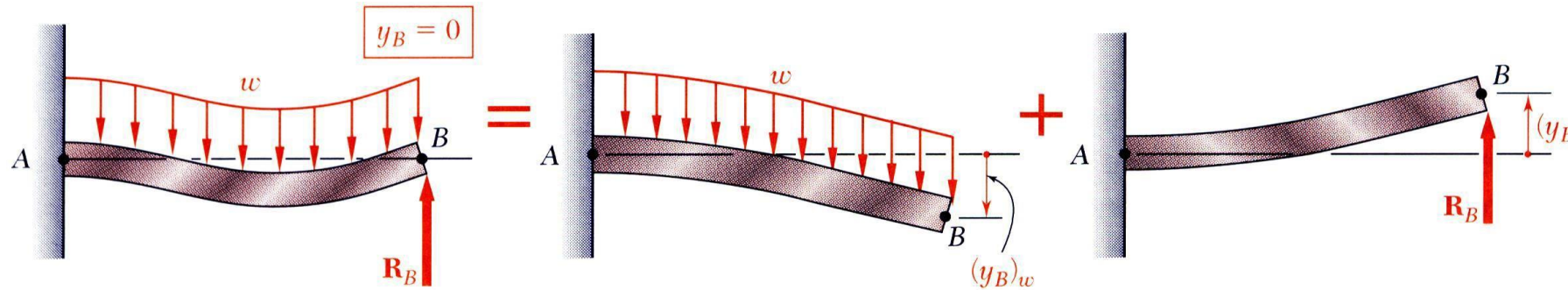
$$y = \frac{w_0}{120EI} \left(-x^5 + 2L^2 x^3 - L^4 x \right)$$

- Differentiate once to find the slope,

$$\theta = \frac{dy}{dx} = \frac{w_0}{120EI} \left(-5x^4 + 6L^2 x^2 - L^4 \right)$$

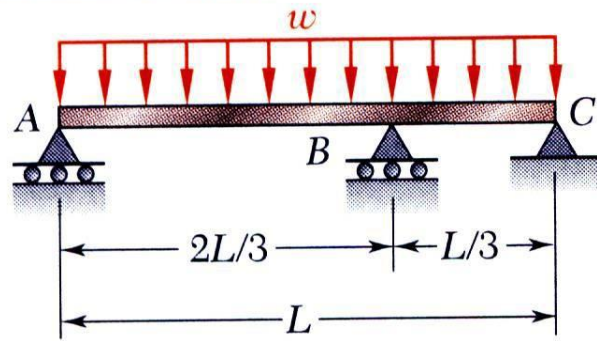
$$\text{at } x = 0, \quad \theta_A = \frac{w_0 L^3}{120EI}$$

Application of Superposition to Statically Indeterminate Beams



- Method of superposition may be applied to determine the reactions at the supports of statically indeterminate beams.
- Designate one of the reactions as redundant and eliminate or modify the support.
- Determine the beam deformation without the redundant support.
- Treat the redundant reaction as an unknown load which, together with the other loads, must produce deformations compatible with the original supports.

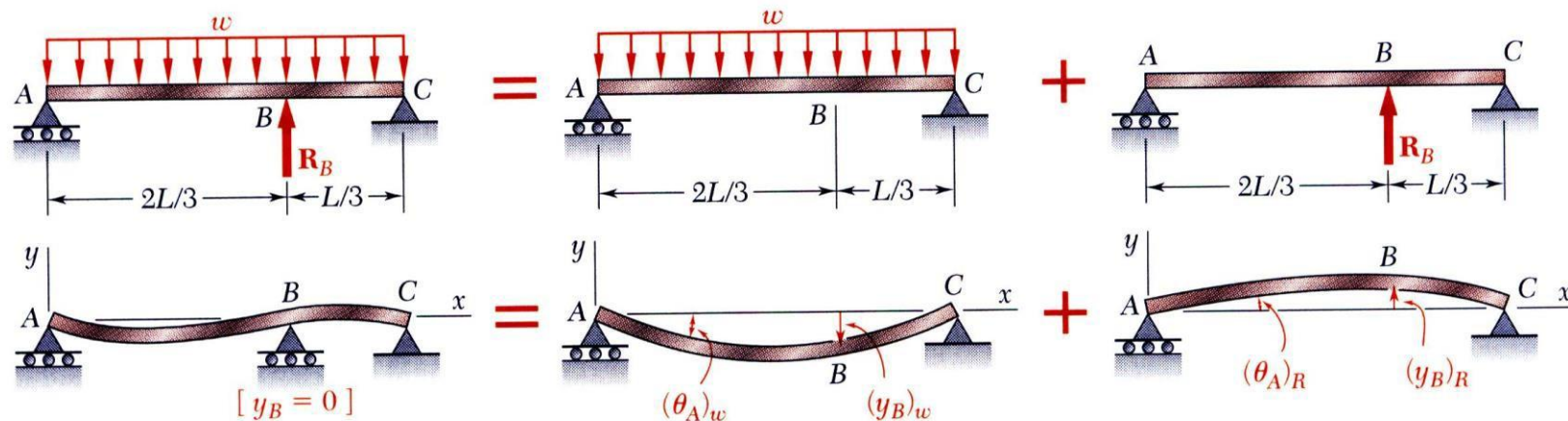
Sample Problem 9.8



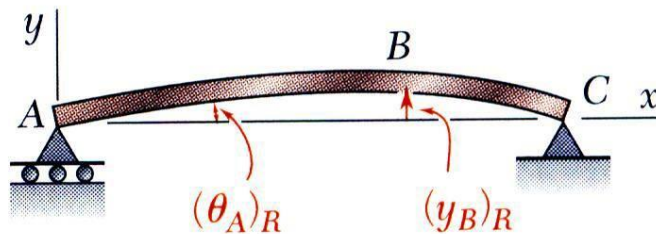
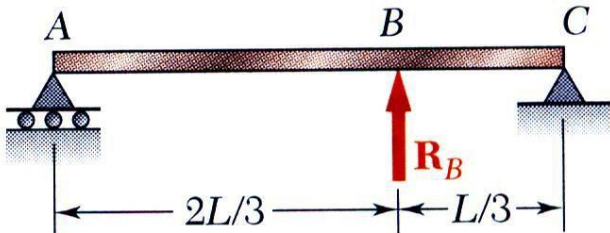
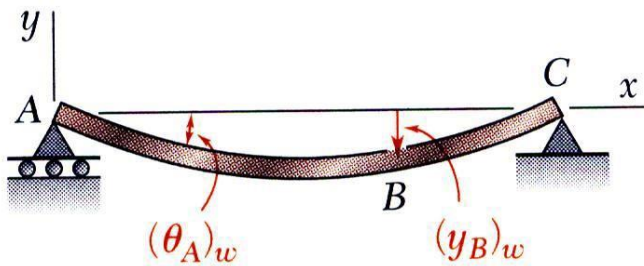
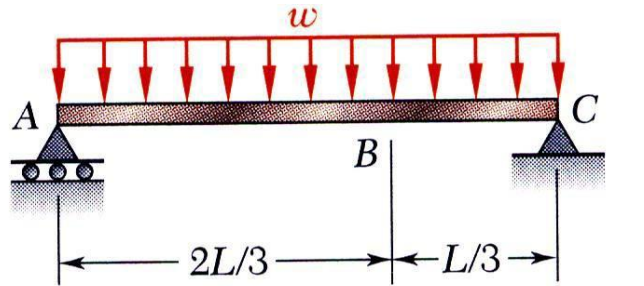
For the uniform beam and loading shown, determine the reaction at each support and the slope at end A .

SOLUTION:

- Release the “redundant” support at B , and find deformation.
- Apply reaction at B as an unknown load to force zero displacement at B .



Sample Problem 9.8



- Distributed Loading:

$$(y_B)_w = -\frac{w}{24EI} \left[\left(\frac{2}{3}L \right)^4 - 2L \left(\frac{2}{3}L \right)^3 + L^3 \left(\frac{2}{3}L \right) \right]$$

$$= -0.01132 \frac{wL^4}{EI}$$

- Redundant Reaction Loading:

$$(y_B)_R = \frac{R_B}{3EI} \left(\frac{2}{3}L \right)^2 \left(\frac{L}{3} \right)^2 = 0.01646 \frac{R_B L^3}{EI}$$

- For compatibility with original supports, $y_B = 0$

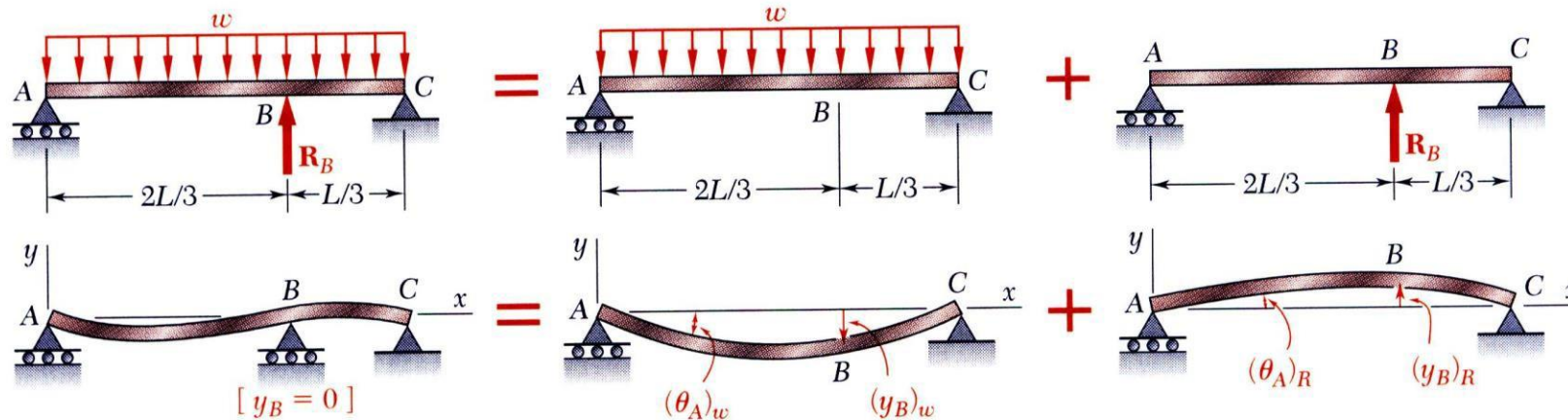
$$0 = (y_B)_w + (y_B)_R = -0.01132 \frac{wL^4}{EI} + 0.01646 \frac{R_B L^3}{EI}$$

$$R_B = 0.688wL \uparrow$$

- From statics,

$$R_A = 0.271wL \uparrow \quad R_C = 0.0413wL \uparrow$$

Sample Problem 9.8



Slope at end A,

$$(\theta_A)_w = -\frac{wL^3}{24EI} = -0.04167 \frac{wL^3}{EI}$$

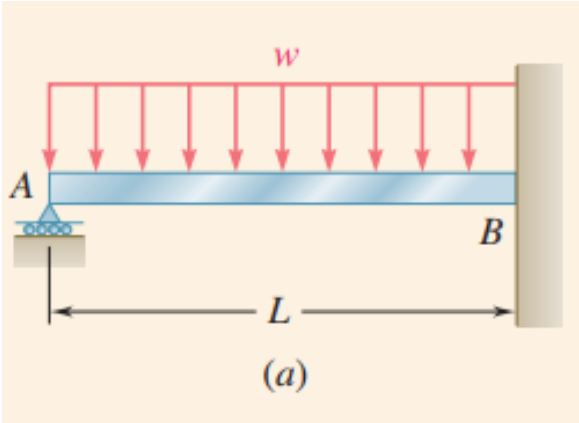
$$(\theta_A)_R = \frac{0.0688wL}{6EI} \left(\frac{L}{3} \right) \left[L^2 - \left(\frac{L}{3} \right)^2 \right] = 0.03398 \frac{wL^3}{EI}$$

$$\theta_A = (\theta_A)_w + (\theta_A)_R = -0.04167 \frac{wL^3}{EI} + 0.03398 \frac{wL^3}{EI}$$

$$\theta_A = -0.00769 \frac{wL^3}{EI}$$

Statically Indeterminate Structures

Castigliano's Theorem



Determine the reactions at the supports for the prismatic beam and load shown

The beam is statically indeterminate to the first degree. The reaction at A is redundant and the beam is released from that support. The reaction R_A is considered to be an unknown load and will be determined under the condition that the deflection y_A at A must be zero.

By Castigliano's theorem

U is the strain energy of the beam under the distributed load and the redundant reaction

$$y_A = \partial U / \partial R_A,$$

$$y_A = \frac{\partial U}{\partial R_A} = \int_0^L \frac{M}{EI} \frac{\partial M}{\partial R_A} dx$$

Bending moment M

$$M = R_A x - \frac{1}{2} w x^2$$

$$\frac{\partial M}{\partial R_A} = x$$

Derivative with respect to R_A

Substituting for M and $\partial M / \partial R_A$

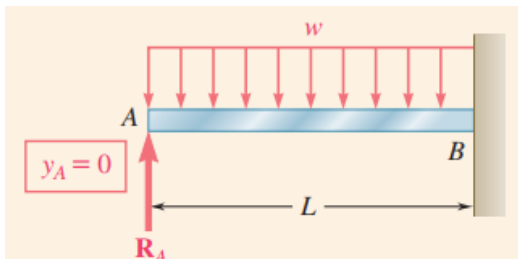
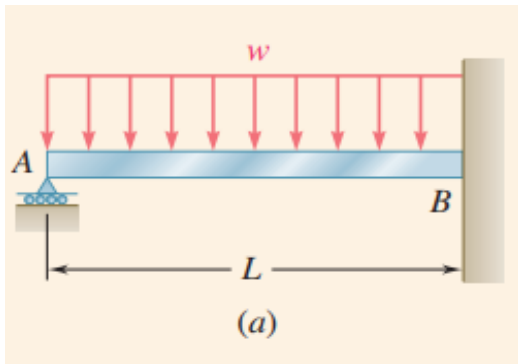
$$y_A = \frac{1}{EI} \int_0^L \left(R_A x^2 - \frac{1}{2} w x^3 \right) dx = \frac{1}{EI} \left(\frac{R_A L^3}{3} - \frac{w L^4}{8} \right)$$

Set $y_A = 0$ and solve for R_A :

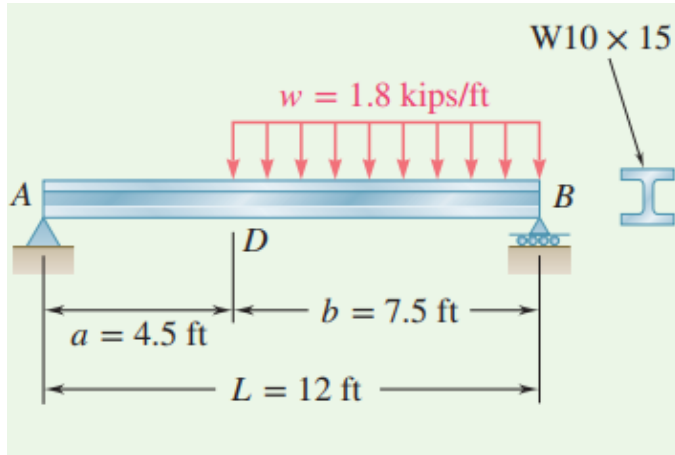
$$R_A = \frac{3}{8} w L \quad \mathbf{R}_A = \frac{3}{8} w L \uparrow$$

From the conditions of equilibrium for the beam, the reaction at B consists of the force and couple:

$$\mathbf{R}_B = \frac{5}{8} w L \uparrow \quad \mathbf{M}_B = \frac{1}{8} w L^2 \curvearrowright$$



Sample Problem 11.6



For the beam and loading shown, determine the deflection at point D .
Use $E = 29 \times 10^6$ psi.

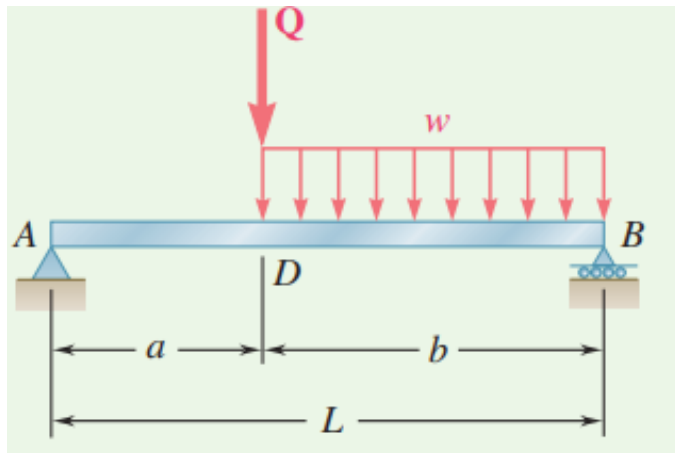
STRATEGY: Add a dummy load associated with the desired vertical deflection at joint D . Use a free-body diagram to determine the reactions due to both the dummy load and the distributed load. The moments in each segment are then written as a function of the coordinate along the beam.

MODELING and ANALYSIS:

Castigliano's Theorem. We introduce a dummy load Q as shown in

Using Castigliano's theorem and noting that EI is constant, write

$$y_D = \int \frac{M}{EI} \left(\frac{\partial M}{\partial Q} \right) dx = \frac{1}{EI} \int M \left(\frac{\partial M}{\partial Q} \right) dx$$



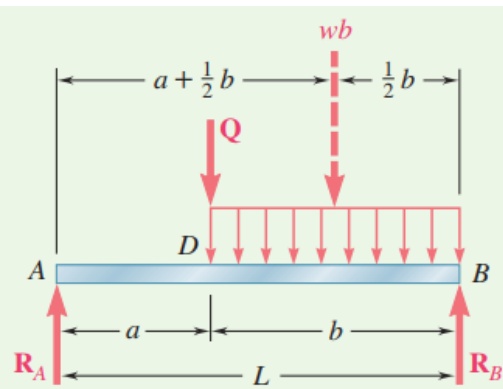


Fig. 2 Free-body diagram of beam.

From A to D

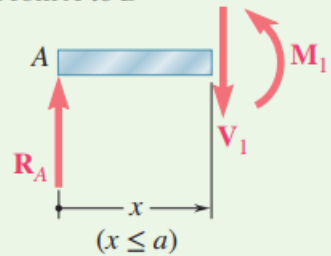


Fig. 3 Free-body diagram of left portion (in AD).

From B to D

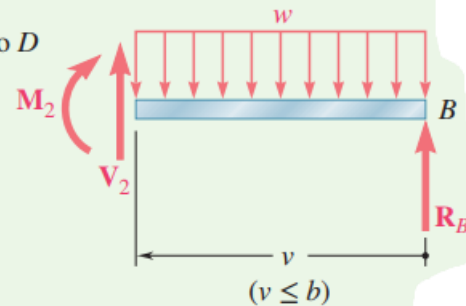


Fig. 4 Free-body diagram of right portion (in BD).

Reactions. Using the free-body diagram of the entire beam (Fig. 2) gives

$$R_A = \frac{wb^2}{2L} + Q\frac{b}{L} \uparrow \quad R_B = \frac{wb(a + \frac{1}{2}b)}{L} + Q\frac{a}{L} \uparrow$$

Portion AD of Beam. Using the free-body diagram shown in Fig. 3,

$$M_1 = R_A x = \left(\frac{wb^2}{2L} + Q\frac{b}{L} \right) x \quad \frac{\partial M_1}{\partial Q} = +\frac{bx}{L}$$

substituting into Eq. (1) and integrating from A to D gives

$$\frac{1}{EI} \int M_1 \frac{\partial M_1}{\partial Q} dx = \frac{1}{EI} \int_0^a R_A x \left(\frac{bx}{L} \right) dx = \frac{R_A a^3 b}{3EI L}$$

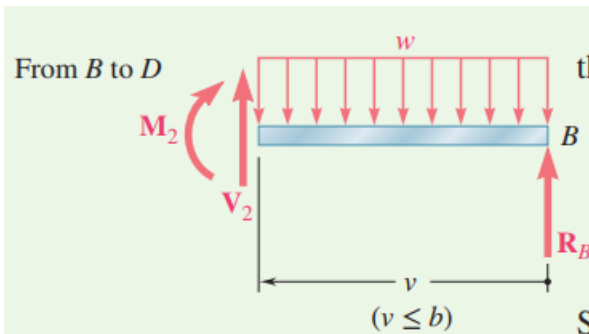


Fig. 4 Free-body diagram of right portion (in BD).

Portion DB of Beam. Using the free-body diagram shown in Fig. 4, the bending moment at a distance v from end B is

$$M_2 = R_B v - \frac{wv^2}{2} = \left[\frac{wb(a + \frac{1}{2}b)}{L} + Q \frac{a}{L} \right] v - \frac{wv^2}{2} \quad \frac{\partial M_2}{\partial Q} = + \frac{av}{L}$$

Substitute into Eq. (1) and integrate from point B (where $v = 0$) to point D (where $v = b$) for

$$\frac{1}{EI} \int M_2 \frac{\partial M_2}{\partial Q} dv = \frac{1}{EI} \int_0^b \left(R_B v - \frac{wv^2}{2} \right) \left(\frac{av}{L} \right) dv = \frac{R_B ab^3}{3EIL} - \frac{wab^4}{8EIL}$$

Substituting for R_B and setting $Q = 0$,

$$\frac{1}{EI} \int M_2 \frac{\partial M_2}{\partial Q} dv = \left[\frac{wb(a + \frac{1}{2}b)}{L} \right] \frac{ab^3}{3EIL} - \frac{wab^4}{8EIL} = \frac{5a^2b^4 + ab^5}{24EIL^2} w \quad (3)$$

Deflection at Point D. Recalling Eqs. (1), (2), and (3),

$$y_D = \frac{wab^3}{24EIL^2} (4a^2 + 5ab + b^2) = \frac{wab^3}{24EIL^2} (4a + b)(a + b) = \frac{wab^3}{24EIL} (4a + b)$$

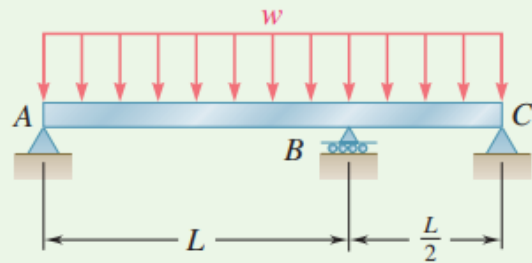
Deflection at Point D. Recalling Eqs. (1), (2), and (3),

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From Appendix E, $I = 68.9 \text{ in}^4$ for a W10 $\times$ 15 beam. Substituting the numerical values for I , w , a , b , and L ,

$$y_D = 0.262 \text{ in.} \downarrow$$

Sample Problem 11.7



For the uniform beam and loading shown, determine the reactions at the supports.

STRATEGY: The beam is indeterminate to the first degree, and we must choose one of the reactions as a redundant. We then use a free-body diagram to solve for the reactions due to the distributed load and the redundant reaction. Using free-body diagrams of the segments, we obtain the moments as a function of the coordinate along the beam. Using Eq. (11.57), we write Castigliano's theorem for deflection associated with the redundant reaction. We set this deflection equal to zero, and solve for the redundant reaction. Equilibrium can then be used to find the other reactions.

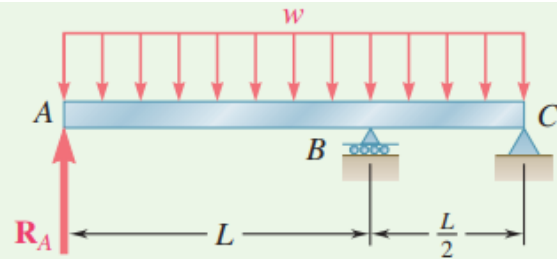


Fig. 1 Released beam, replacing support at A with redundant reaction R_A .

MODELING and ANALYSIS:

Castigliano's Theorem. Choose the reaction $\mathbf{R}_A$ as the redundant one (Fig. 1). Using Castigliano's theorem, determine the deflection at A due to the combined action of $\mathbf{R}_A$ and the distributed load. Since EI is constant,

$$y_A = \int \frac{M}{EI} \left(\frac{\partial M}{\partial R_A} \right) dx = \frac{1}{EI} \int M \frac{\partial M}{\partial R_A} dx \quad (1)$$

The integration will be performed separately for portions AB and BC of the beam. $\mathbf{R}_A$ is then obtained by setting y_A equal to zero.

Free Body: Entire Beam. Using Fig. 2, the reactions at B and C in terms of R_A and the distributed load are

$$R_B = \frac{9}{4} wL - 3R_A \quad R_C = 2R_A - \frac{3}{4} wL \quad (2)$$

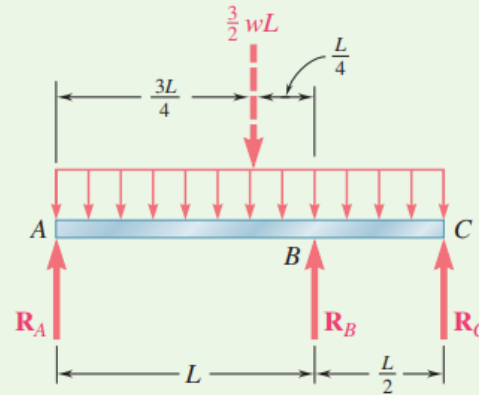


Fig. 2 Free-body diagram of beam.

From A to B

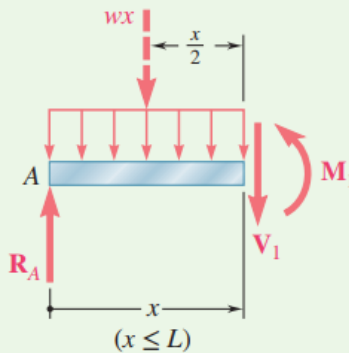


Fig. 3 Free-body diagram of left portion showing internal shear and moment.

Portion AB of Beam. Using the free-body diagram shown in Fig. 3, find

$$M_1 = R_A x - \frac{wx^2}{2} \quad \frac{\partial M_1}{\partial R_A} = x$$

Substituting into Eq. (1) and integrating from A to B gives

$$\frac{1}{EI} \int M_1 \frac{\partial M}{\partial R_A} dx = \frac{1}{EI} \int_0^L \left(R_A x^2 - \frac{wx^3}{2} \right) dx = \frac{1}{EI} \left(\frac{R_A L^3}{3} - \frac{wL^4}{8} \right) \quad (3)$$

Portion BC of Beam. Using the free-body diagram shown in Fig. 4, find

From C to B

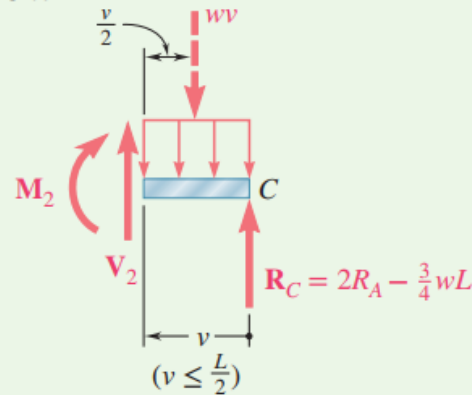


Fig. 4 Free-body diagram of right portion showing internal shear and moment.

$$M_2 = \left(2R_A - \frac{3}{4} wL \right) v - \frac{wv^2}{2} \quad \frac{\partial M_2}{\partial R_A} = 2v$$

Substituting into Eq. (1) and integrating from C (where $v = 0$) to B (where $v = \frac{1}{2}L$) gives

$$\begin{aligned} \frac{1}{EI} \int M_2 \frac{\partial M_2}{\partial R_A} dv &= \frac{1}{EI} \int_0^{L/2} \left(4R_A v^2 - \frac{3}{2} wL v^2 - wv^3 \right) dv \\ &= \frac{1}{EI} \left(\frac{R_A L^3}{6} - \frac{wL^4}{16} - \frac{wL^4}{64} \right) = \frac{1}{EI} \left(\frac{R_A L^3}{6} - \frac{5wL^4}{64} \right) \end{aligned} \quad (4)$$

Reaction at A. Adding the expressions from Eqs. (3) and (4), we obtain y_A and set it equal to zero:

$$y_A = \frac{1}{EI} \left(\frac{R_A L^3}{3} - \frac{wL^4}{8} \right) + \frac{1}{EI} \left(\frac{R_A L^3}{6} - \frac{5wL^4}{64} \right) = 0$$

Thus, $R_A = \frac{13}{32} wL$ $R_A = \frac{13}{32} wL \uparrow$ ◀

Reactions at B and C. Substituting for R_A into Eq. (2), we obtain

$$R_B = \frac{33}{32} wL \uparrow \quad R_C = \frac{wL}{16} \uparrow \quad \text{◀}$$



Thank you

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